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Version 2.0-alpha



This version of the package is already quite powerful, but it is still in the proof-of-concept phase. There are bugs and the documentation is incomplete. Several planned features have not been implemented and some things that have been implemented are very rough. Everything is subject to change.

The package will eventually be released on ctan. Comments and suggestions are welcome.

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This package provides commands for drawing Young diagrams, tableaux, abacuses and other common objects used in algebraic combinatorics and representation theory. The commands are designed to be easy to use and easy to read. The literature contains many embellishments of the basic diagrams, so we provide mechanisms to customise and style these diagrams.

The package provides basic commands for drawing these objects, together with styling options and a key-value interface for adding extra features. Under the hood, everything is drawn using `TikZ`. It is possible to use all of our commands either as standalone commands, or as components of a `tikzpicture` environment. This makes it possible to use these commands inside more complicated diagrams.

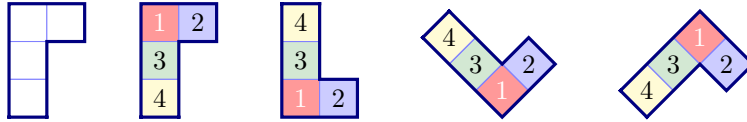
1. YOUNG DIAGRAMS AND TABLEAUX

Partitions, which are weakly decreasing sequences of non-negative integers, are fundamental objects in algebraic combinatorics and representation theory. Rather than working with sequences of integers, it is often useful to visualise partitions as Young *diagrams*, which are arrays of *boxes*, or *nodes*, in the plane. A *tableau* is a (Young) diagram where the boxes are labelled using some alphabet. Equivalently, a diagram is a tableau with empty labels.

For example, $\lambda = (4, 3, 3, 2, 0, 0, \dots)$ is a partition of 12, since the entries sum to 12. It is customary to omit the zeros when writing partitions and to use exponents for repeated parts, so we also write $\lambda = (4, 3^2, 2)$. Many authors identify a partition $\lambda = (\lambda_1, \lambda_2, \dots)$ with its *diagram*, which is the set

$$[\lambda] = \{ (r, c) \mid 1 \leq c \leq \lambda_r \text{ for } r \geq 1 \}.$$

The diagram of λ is visualised as a justified array of boxes in the plane, where the first row has 4 boxes, the next two rows have 3 boxes and the last row has 2 boxes. The *shape* of a diagram is the partition that gives the number of boxes in each row. Here are some examples of the diagrams and tableaux of shape $(2, 1^2)$ that can be drawn using `aTableau`:



This choice of colouring is gratuitous, but it emphasises that `aTableau` allows you to add style (and, in particular, colour), to every box in these diagrams. This example also shows that `aTableau` supports the four different conventions used for drawing Young diagrams and tableaux. The following table shows how to draw the Young diagram of shape $\lambda = (4, 3^2, 2)$ using the four tableaux conventions supported by `aTableau`:

Convention	<code>aTableau</code> command	Diagram
English (the default)	<code>\Tableau[english]{1234,567,89{10},{11}{12}}</code>	
French	<code>\Tableau[french]{1234,567,89{10},{11}{12}}</code>	
Ukrainian ¹	<code>\Tableau[ukrainian]{1234,567,89{10},{11}{12}}</code>	
Australian ²	<code>\Tableau[australian]{1234,567,89{10},{11}{12}}</code>	

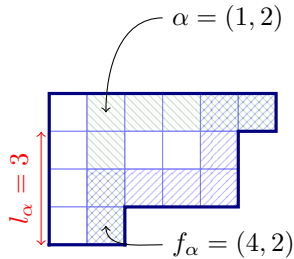
The different conventions for diagrams, and tableaux, are listed in order of their popularity in the literature. By default, the *English* convention is used, so `\Diagram{4,3,3,2}` produces the first diagram above. These four conventions can be applied in the same way to the `\Tableau` command. You can change the default convention using the `\aTabSet` command.

This table shows how the commands provided by `aTableau` work: there are basic commands for drawing diagrams, and these diagrams can be embellished using optional arguments and a key-value interface. In addition, as the following example shows, by supplying (optional) Cartesian coordinates, these diagrams can be used inside a `tikzpicture` environment to create more exotic pictures:

```
% \usetikzlibrary{patterns}
\begin{tikzpicture}[
  B/.style={pattern=north east lines, pattern color=blue,opacity=0.5},
  Y/.style={pattern=north west lines, pattern color=ForestGreen,opacity=0.5},
]
\Diagram(0,0)[ribbons={{(Y)16ccccrrr, (B)16crrcccr}}{6,5^2,2}
\draw[->](1.5, 1)node[right]{ $\alpha=(1,2)$ } to [out=180,in=90] (B-1-2.base);
\draw[->](1.5,-2)node[right]{$f_\alpha=(4,2)$ } to [out=180,in=270] (B-4-2.base);
\draw[red,<->]([xshift=-1mm]B-1-1.south west)
--node[rotate=90,above]{$1_\alpha=3$}([xshift=-1mm]B-4-1.south west);
\end{tikzpicture}
```

¹Some authors call this the Russian convention

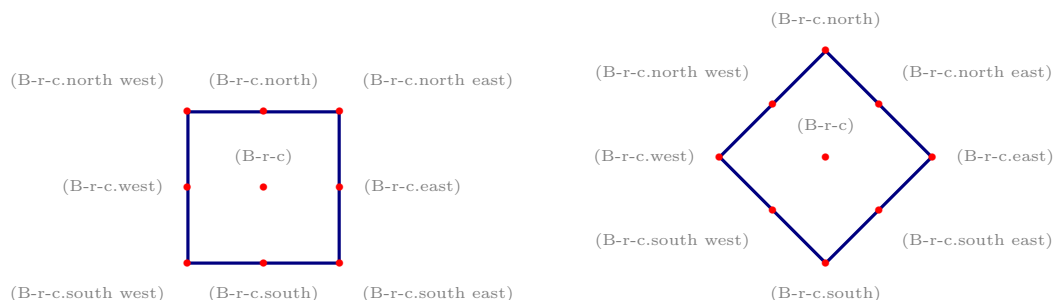
²I have not seen this convention in the literature, but this name for the convention seems apt.



As example shows, partitions can either be typed out in full or you can use exponential notation, which is often shorter and easier to read.

Most of the commands in this example are `TikZ` commands, but most of the picture is drawn by the `\Diagram` command. The top of the example defines two `TikZ` styles, `B` and `Y`, which are used in the ribbons. The Young diagram is placed at position (0,0) by the `\Diagram` command. The three `\draw` commands in this example are used to add the three arrows to the picture. You should consult the very readable `TikZ` manual for an explanation of how these commands work.

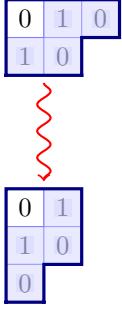
From the viewpoint of `aTableau`, the most important point of this example is that the `\Diagram` command can be used inside a `tikzpicture` environment because it is equipped with the Cartesian coordinate (0,0). The example also uses the four named coordinates, (B-1-1), (B-1-2), (B-4-1), and (B-4-2). When `aTableau` draws a tableau, or a diagram, it creates `TikZ` named coordinates for each of the boxes in the diagram, so that (B-r-c) refers to the box in row r and column c. Giving finer control, the following standard `TikZ anchors` are available:



Here, r and c should be replaced by the row and column index of a box in the tableau, where these anchors are only defined for boxes in the tableau. These *anchors* are a standard part of **TikZ**. If you change the shape of the node then the available anchors may change; for more information see the **TikZ** manual. You can change the B in the name of these anchors to anything you like using the **prefix** option.

The previous example also shows that the `\Diagram` command accepts a `ribbons` option, which adds *ribbons* to the diagram. Moreover, these ribbons can be given `TikZ` styling. All of the options to these commands are described in the following sections.

In the example above the Cartesian coordinate $(0,0)$ is necessary because the `\Diagram` command is embedded inside a `tikzpicture` environment, but the coordinate itself is not important in the sense that changing the coordinate will not change the picture. Cartesian coordinates become more meaningful when there are more components inside the `tikzpicture` environment. Consider:

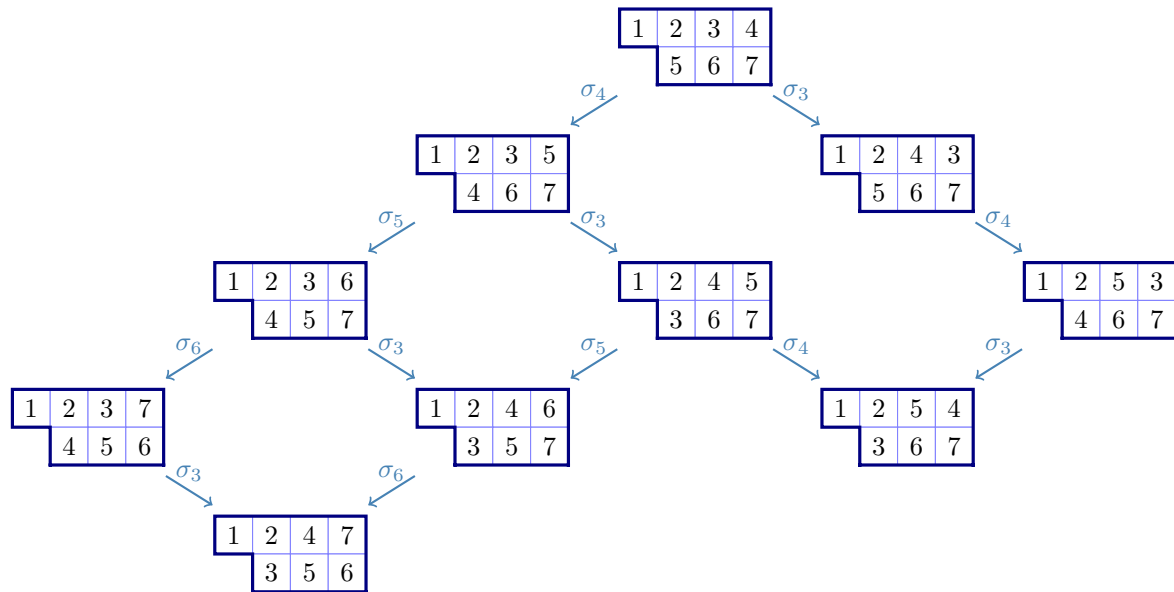


```
% \usetikzlibrary{decorations.pathmorphing}
\begin{tikzpicture}[wave/.style={thick,red,->,decorate,decoration=snake}]
  \aTabSet{ribbonstyle={fill=blue!20, opacity=0.4}}
  \Diagram(0,0)[e=2,show=residues,ribbons={13crc},prefix=A]{3,2}
  \Diagram(0,-2.5)[e=2,show=residues,ribbons={12rcr},prefix=B]{2^2,1}
  \draw[wave](0.5,-1.1)--(0.5,-2.4);
\end{tikzpicture}
```

The following sections describe the commands defined by the package and how to use them. We start with the `\Tableau` command because all of the diagram and tableau commands, except for `\RibbonTableau`, are derived from `\Tableau`.

Finally, the condition that coordinates are required only when these diagrams are being used inside a `tikzpicture` environment needs to be interpreted appropriately. For example, coordinates are not required when the diagrams are inside nodes in these diagrams:

```
% \usetikzlibrary{matrix}
\begin{tikzpicture}[arr/.style={->,SteelBlue, thick}]
  \matrix (M)[matrix of nodes,row sep=4mm,column sep=4mm]{
    & & & \ShiftedTableau{1234,567} & \\
    & & \ShiftedTableau{1235,467} & & \\
    & & \ShiftedTableau{1243,567} & & \\
    & \ShiftedTableau{1236,457} & & & \\
    & & \ShiftedTableau{1245,367} & & \\
    & & \ShiftedTableau{1253,467} & & \\
    \ShiftedTableau{1237,456} & & & & \\
    & & \ShiftedTableau{1246,357} & & \\
    & & \ShiftedTableau{1254,367} & & \\
    & \ShiftedTableau{1247,356} & & & \\
  };
  \draw[arr] (M-1-4)--node[above]{$\sigma_4$}(M-2-3);
  \draw[arr] (M-1-4)--node[above]{$\sigma_3$}(M-2-5);
  \draw[arr] (M-2-3)--node[above]{$\sigma_5$}(M-3-2);
  \draw[arr] (M-2-3)--node[above]{$\sigma_3$}(M-3-4);
  \draw[arr] (M-2-5)--node[above]{$\sigma_4$}(M-3-6);
  \draw[arr] (M-3-2)--node[above]{$\sigma_6$}(M-4-1);
  \draw[arr] (M-3-2)--node[above]{$\sigma_3$}(M-4-3);
  \draw[arr] (M-3-4)--node[above]{$\sigma_5$}(M-4-3);
  \draw[arr] (M-3-4)--node[above]{$\sigma_4$}(M-4-5);
  \draw[arr] (M-3-6)--node[above]{$\sigma_3$}(M-4-5);
  \draw[arr] (M-4-1)--node[above]{$\sigma_3$}(M-5-2);
  \draw[arr] (M-4-3)--node[above]{$\sigma_6$}(M-5-2);
\end{tikzpicture}
```



1.1. **Tableaux.** This section describes the `\Tableau` command, which draws tableaux or labelled diagrams. The syntax for this command is:

`\Tableau (x,y) [options] {tableau entries}`

(x,y) :
The Cartesian coordinates (x,y) are needed if and only if the tableau is inside a `tikzpicture` environment.

options :
Optional arguments, described below.

tableau entries :
The `tableau entries` are given as a comma separated list, with commas separating rows and each *token*, or braced-group of tokens, being in a separate column. As we explain below, each tableau entry can be prefixed by (optional) **TikZ** styling commands.

We show by example how to use the `\Tableau` command, starting with basic usage and finishing with style. Inside `\Tableau`, the rows are separated by commas, with the column entries given by the “letters” in the entry for each row:

1	2	3	4
5	6		
7			
8			

α	γ	δ	q
x	y	η	
λ	μ		
π	$\sum$		

```
\Tableau{ 1234, 56, 7, 8 }
\Tableau{ \alpha\gamma\delta q,
          xy\eta,
          \lambda\mu,
          \pi\sum }
```

You can put all of the tableau specifications on one line, without any spaces. Alternatively, as I do, you add spaces and line breaks between the tableau entries to improve readability (although, you cannot have blank lines inside a `\Tableau` command). As this example suggests, by default, the entries of a tableau are typeset as mathematics, but this can be changed.

As the second example above shows, the tableau entries are *not* just individual characters: they also include $\text{\TeX}$ commands. Tableaux frequently contain entries that are not single characters, or given by commands. Such entries can be inserted in a tableau by enclosing them in braces:

1	3	10	11	$2x$
2	x^2			

```
\Tableau{ 13{10}{11}{2x}, 2{x^2} }
```

There is one additional caveat for braced-entries: any *singleton* braced entry `{...}` in a row needs to be double-braced. The following example shows the difference between single and double-bracing of tableau entries.

1	3	4	5
8	10		
1	1		
1	2		

1	3	4	5
8	10		
11			
12			

```
\Tableau{ 1345, 8{10}, {11}, {12} }
\Tableau{ 1345, 8{10}, {{11}}, {{12}} }
```

The need for double-bracing of singleton entries is a consequence of how L^AT_EX3 detects braced commas in comma separated lists. Double-bracing is only necessary when you have a single entry in a column that requires braces, which is only if the row contains a single column. The `{11}` and `{12}` in the first tableau are both treated as spanning two columns in the tableau because they are not doubled-braced; in contrast, they occupy one column in the second tableau because they are double-braced.

Adding style. The `aTableau` package provides several different ways to add `TikZ` style commands to the entries in tableaux, and to boxes in diagrams. To do this you simply add style commands as a prefix. This is made possible by adding style prefixes to the tableau entries. To take advantage of these styles you will need at least rudimentary knowledge of how `TikZ` styles work. The examples in this manual give a good indication of what is possible. For anything more exotic, please consult the `TikZ` manual.

The simplest way to change the style of an entry in a tableau is to add `*`-prefixes to the entry in the tableau specifications.

1	2	3	4	5
6	7	8	9	
10	11	12		
13				

```
\Tableau{ 1*2*3*4*5, 6*789, {10}*{11}{12}, {{13}} }
```

By default, the `*`-entries are given the `TikZ` styling `fill=blue!20`. You can change the default `*`-style using the `starstyle` key:

1	2	3	4	5
6	7	8	9	
10	11	12		
13				

```
\Tableau[starstyle={fill=red!20,draw=red,thick}]
{ 1*2*3*4*5, 6*789, {10}*{11}{12}, {{13}} }
```

Notice that the `*`-styling takes precedence over the default styling of the surrounding boxes. This is because the boxes with the default styling are placed first, in the order that they are entered, after which the boxes with styling are placed.

The `*`-syntax is convenient if you want to emphasise some of the boxes in a tableau, using a single style. In fact, it is possible to give every box in the tableau its own style by putting style specifications inside square brackets, `[...]`, before the corresponding letter in the tableau specification. You can mix the `*`-syntax and the `[...]`-style syntax for different entries, although you can only apply one of these style settings to any given node.

13				
10	11	12		
6	7	8	9	
1		3		

```
\Tableau[french]{
  1 [blue!20]2 [circle]3 [red]4 [cyan]5,
  6 *7 8 9,
  {10} *{11} {12},
  {{13}} }
```

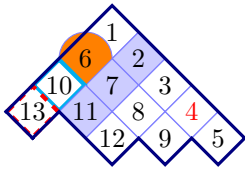
It is not necessary to add spaces between the entries like this. This example uses spaces only because this (hopefully) makes it clear which style specification applies to which box in the tableau.

Omitting some technicalities, each box in a tableau is constructed as a `TikZ` node, so any `TikZ` style settings for a node can be used. This example shows that some precision is needed when changing the style because simply giving a colour will change both the colour used to fill the box and the text colour, which is why some of tableau entries above appear to be blank. If you are already familiar with `TikZ` then you will know how to do this. If not, then here is a short list of useful style specifications for `TikZ` nodes:

Style	Meaning
<code>text = <colour></code>	Sets the text colour in the box
<code>fill = <colour></code>	Sets the fill, or background, colour of the box
<code>draw = <colour></code>	Sets the boundary colour of the box

In the style settings, any valid L^AT_EX colour name can be used (see the `xcolor` manual). In addition, `thin`, `thick`, `very thick`, `ultra thick`, `dashed`, `dotted`, ... change the border of the corresponding box.

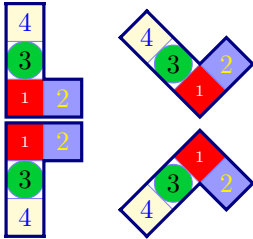
If you want to apply “complicated” style settings that consist of a comma separated list of style settings, such as `draw=cyan,ultra thick`, then the entire style has to be put inside matching square and curly braces `[{...}]`. This is necessary because otherwise L^AT_EX gets confused since commas are used to separate the rows of the tableau.



```
\Tableau[australian]{
  1 [fill=blue!20]2 3 [text=red]4 5,
  [{circle,fill=orange}]6 *7 8 9,
  [{draw=cyan,ultra thick}]{10} *{11} {12},
  [{dashed,draw=red,ultra thick}]{13} }
```

This last example shows that it is not necessary to double-brace singleton entries when they have a style specification. As this example shows, modifying the borders of a box does not add much emphasis, so use this sparingly. In addition, the orange circle in this example looks too big, but it is what we asked for because its diameter matches the horizontal width of the box. The next example shows that we get a better result for *Australian* and *Ukrainian* tableaux if we add a *minimum size* specification.

For complicated styles, or styles that are used often, we recommend defining the style outside of the `\Tableau` command by using the `\tikzset` command to define the style:



```
\tikzset{
  B/.style={fill=blue!40, text=yellow},
  G/.style={fill=green!80!blue,circle,minimum size=5mm},
  R/.style={fill=red,text=white, font=\tiny},
  Y/.style={fill=yellow!20, text=blue}
}
\Tableau[french]{[R]1[B]2,[G]3,[Y]4}
\Tableau[ukrainian]{[R]1[B]2,[G]3,[Y]4}
\Tableau[english]{[R]1[B]2,[G]3,[Y]4}
\Tableau[australian]{[R]1[B]2,[G]3,[Y]4}
```

In this example, the `TikZ` styles `B`, `G`, `R`, and `Y` are being applied to the tableaux entries labelled 1, 2, 3, and 4, respectively. The colours used here, and throughout this manual, are for demonstration purposes only: I recommend against such garish choices in any self-respecting document!

\Tableau options. The optional `*-styles` and `[...]-styles` are the main mechanism for styling individual nodes in a tableau. Using a key-value syntax, you can change other aspects of the tableaux produced by the `\Tableau` command. The rest of this section describes these keys, their default values, and gives examples of how to use them.

The options described below can be used with the `\Tableau` command by placing them as a comma separated list inside square brackets:

`\Tableau[options]{tableau entries}` or `\Tableau(x,y)[options]{tableau entries}`.

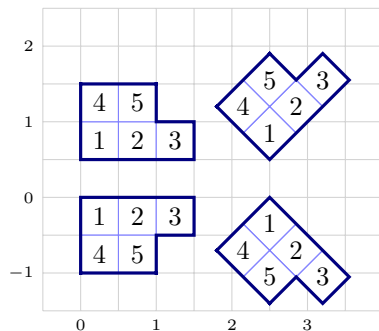
The options can appear in any order, with later options having precedence over earlier ones.

When used inside a `\Tableau` command, the options only affect that particular tableau. Most of these options can also be used in the `\aTabSet` command, to change the default settings of all following tableaux in the same L^AT_EX group. For example, if you put the command `\aTabSet[ukrainian]` in the preamble of your document then, by default, latter tableaux and Young diagrams will be drawn using the Ukrainian convention.

(x, y)

The Cartesian coordinates (x, y)

The (x, y) -coordinates are needed if, and only if, `\Tableau` command is used inside a `tikzpicture` environment, in which case (x, y) gives the coordinates of the “outside corner” of the box in row 1 and column 1 of the tableau. The following example shows how the tableaux are placed using coordinates inside a `tikzpicture` environment. The example also shows that, by default, the tableau boxes are half a unit wide.



```
\begin{tikzpicture}[addgrid]
  \Tableau(0,0) [english] {123,45}
  \Tableau(0,0.5) [french] {123,45}
  \Tableau(2.5,0.5) [ukrainian] {123,45}
  \Tableau(2.5,0) [australian] {123,45}
\end{tikzpicture}
```

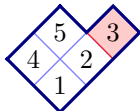
english (default: **english**)

french

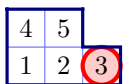
ukrainian

australian

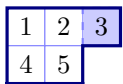
These four options change the convention, or orientation that is used to draw the tableaux. Rather than define this precisely, we use the tired and true method in combinatorics of defining by example.



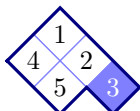
```
\tikzset{R/.style={fill=red!20,draw=red}}
\Tableau[ukrainian]{12[R]3,45}
```



```
\tikzset{R/.style={fill=red!20, circle, draw=red, ultra
thick}}
\Tableau[french]{12[R]3,45}
```



```
\tikzset{B/.style={fill=blue!20, dashed, thick}}
\Tableau[english]{12[{B, thick}]3,45}
```



```
\tikzset{B/.style={fill=blue!50, thick, text=white}}
\Tableau[australian]{12[{B, thick}]3,45}
```

By default, the english convention is used. This can be changed using `\aTabSet`. The capitalised names, **Australian**, **English**, **French** and **Ukrainian**, are also recognised. If, for example, your document is only going to french tableaux and diagrams, then you can specify this when you load the package using `\usepackage[french]{atableau}`.

The `\Tableau` settings used as options to the `\usepackage` statement change the default settings throughout the document.

align (default: **center**)

[accepts: centre/bottom/top]

aTableau

By default, tableaux are centred vertically. This can be changed using the `align` key.

1	2	3
4	5	5
6	7	4

1	2	3
4	5	5

```
\Tableau{123,455,674}
\Tableau{123,455}
```

1	2	3
4	5	5
6	7	4

1	2	3
4	5	5

```
\Tableau[align=top]{123,455,674}
\Tableau[align=top]{123,455}
```

1	2	3
4	5	5
6	7	4

1	2	3
4	5	5

```
\Tableau[align=bottom]{123,455,674}
\Tableau[align=bottom]{123,455}
```

1	2	3
4	5	5
6	7	4

1	2	3
4	5	5

```
\Tableau[align=centre]{123,455,674}
\Tableau[align=centre]{123,455}
```

It is not necessary to have the same `align` keys on each tableau, but this one way to ensure that you get the desired result. Here and elsewhere, American spelling variations are also accepted.

conjugate (default: `false`)

[accepts: `false/true`]

Draws the *conjugate* tableau, where the rows and columns are swapped.

1	4	6
2	5	7
3		

```
\Tableau[conjugate]{123,45,67}
```

prefix (default: `B`)

[accepts: `string/character`]

By default, the boxes can be referenced using the node names (B-1-1), (B-1-2), (B-1-3), ..., with (B-*r*-*c*) referring to the node in row *r* and column *c*. This feature is most useful when you have several tableaux inside a `tikzpicture` environment because if you only have one tableau, then it is unlikely that you need to change the default prefix.

tikzpicture

[accepts: `TikZ` style commands]

Use this command to add an optional argument to the underlying `tikzpicture` environment that contains the tableau. The `tikzpicture` option is ignored when the `\Tableau` command is equipped with (*x*,*y*)-coordinates.

13	11	12	9	5
10	7	8	4	
6	3	2		
1				

```
\Tableau[ukrainian, prefix=A,
tikzpicture={remember picture}]
{ 1[fill=blue!20]23[text=red]45,
6*78[{draw=cyan,ultra thick}]9,
[{\circle,fill=orange}]{10}*{11}{12},
[{\dashed,draw=red,ultra thick}]{13}
}
```

aTableau

Since `prefix=A`, the nodes in this tableau are referenced as (A-1-1), (A-1-2), (A-1-3), As the example also uses `tikzpicture=remember picture`, we can reference the node names in last tableau from other `tikzpicture` environments. For example, we can do things like this:

This circle is too big!
Use `minimum size=5mm`

```
\tikz[remember picture, overlay]
\draw[very thick, ->,red]
(0,0) node[right,align=left]
{This circle is too big!\Use minimum size=5mm}
to [out=135, in=225](A-3-1);
```

As a second example, we can use `tikzpicture=rotate=30` to try to rotate a tableau by 90 degrees with:

3	
2	5
1	4

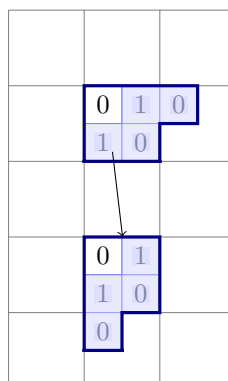
```
\Tableau[tikzpicture={rotate=90}]{123,45}
\begin{tikzpicture}[rotate=90]
\Tableau(0,0){123,45}
\end{tikzpicture}
```

The two tableaux constructions in this example are equivalent. An oddity with both of these examples is that the numbers have not been rotated, which is because *rotate* does not affect component objects, such as nodes, inside a **TikZ** picture. You can rotate everything in the picture by using *transform canvas*:

3	
2	5
1	4

```
\Tableau[tikzpicture={transform canvas={rotate=90}}]{123,45}
```

```
% \usetikzlibrary{decorations.pathmorphing}
\begin{tikzpicture}[wave/.style={thick,red,->,decorate,decoration=snake}]
\TabSet{ribbonstyle={fill=blue!20, opacity=0.4}}
\draw[help lines](-1,-4) grid (2,1);
\Diagram(0,0)[e=2,show=residues,ribbons={13crc},prefix=A]{3,2}
\Diagram(0,-2)[e=2,show=residues,ribbons={12rcr},prefix=B]{2^2,1}
\draw[->](A-2-1.south east)--(B-1-1.north east);
\end{tikzpicture}
```



tikz before
tikz after

[accepts: **TikZ** commands]
[accepts: **TikZ** commands]

These commands inject **TikZ** code, before and after the `\Tableau` command, , respectively, into the underlying `tikzpicture` environment where the tableau is constructed.

	1	3	4	
	5	6	7	
	8	9		

```
\Tableau[
  tikz before={\draw[help lines,step=0.5](-0.5,-2)grid(2,0.5);},
  tikz after={\draw[ultra thick,red]
    (B-1-3.north east)--(B-1-3.south east)
    --(B-2-1.north east)--(B-3-1.south east)
    --(B-3-1.south west);
  }]{134,567,89}
```

Both before and after hooks are provided because the order in which you place **TikZ** commands can affect the drawing, primarily because later commands are drawn over the top of earlier ones.

inner wall
outer wall

inner style
outer style

fill

text

font

shape

math nodes (default: **true**) [accepts: true/false]
text nodes (default: **false**) [accepts: true/false]

node height
node width

ribbons [accepts: a comma separated list of ribbon specifiers]
snobs [accepts: a comma separated list of ribbon specifiers]
ribbonstyle (default: **fill=blue!20**) [accepts: **TikZ** style commands]

Use the **ribbons** key to add a comma separated list of ribbons to your tableau. The syntax for adding ribbons is described in [subsection 1.6](#), which describes the `\RibbonTableau` command. The basic idea is that you specify the location of the head of the ribbon, which has minimum row index and maximum column index, and then gives a sequence of **r**'s and **c**'s depending on whether the ribbon moves along a row or column – and, of course, you can add style. Use the **ribbonstyle** option to change the default style of the ribbons. For example, noting that a box is a ribbon of length 1, the following code uses ribbons of length one (that is, nodes), to highlight the addable nodes of this tableau.

aTableau

1	2	3	4	
5	6	7		
8	9	10		
11				

```
\Tableau[starstyle={fill=green!80!blue,text=white},
ribbonstyle={fill=yellow},
ribbons={15,24,42,51},
]{123*4,567,89*{10},*{11}}
```

The ribbons are always assumed to be contained inside the diagram of the tableau, so they will not change the boundary of the tableau. Use `snoobs` to add ribbons to the tableau that are placed *after* the border of the diagram has been drawn. (Writing *ribbons* backwards gives *snoobs*.) The default style for a `snoobs` ribbon is given by `ribbonstyle`.

1	2	3	4	
5	6	7		
8	9	10		
11				

```
\Tableau[starstyle={fill=green!80!blue,text=white},
ribbonstyle={fill=yellow},
snoobs={15,24,42,51},
]{123*4,567,89*{10},*{11}}
```

The difference between these two pictures is that in the second picture the nodes in the `snoobs` are drawn over the tableau boundary, causing the boundary to disappear for these nodes. For this reason, in most cases you should use `ribbons`, and avoid `snoobs`. I do not know of a real example where I would use `snoobs`, so I would be pleased to hear of any example where it is necessary.

star style

box height (default: 0.5)

a decimal number (in centimetres) **box width** (default: 0.5)

a decimal number (in centimetres) **scale** (default: 1)

[accepts: a decimal number]

xscale (default: 1)

[accepts: a decimal number]

yscale (default: 1)

[accepts: a decimal number]

By design, boxes in tableaux do not automatically resize to fit their contents. For example, consider:

1	3	10	11	$1+x$
2	$1+x$			

```
\Tableau{ 13{10}{11}{1+x}, 2{1+x} }
```

This is unlikely to be the desired output! The options `scale`, `xscale` and `yscale` can be used to rescale the boxes in tableaux and diagrams. The names are slightly misleading because `xscale` rescales in the direction of increasing column index and `yscale` rescales in the direction of increasing row index.

1	3	10	11	$1+x$
2	$1+x$			

```
\Tableau[scale=2]{ 13{10}{11}{1+x}, 2{1+x} }
```

1	3	10	11	$1+x$
2	$1+x$			

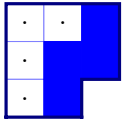
```
\Tableau[xscale=2]{ 13{10}{11}{1+x}, 2{1+x} }
```

1	3	10	11	$1+x$
2	$1+x$			

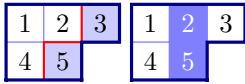
```
\Tableau[ukrainian, xscale=2]{ 13{10}{11}{1+x}, 2{1+x} }
```



The options, `scale`, `xscale`, and `yscale`, affect all `aTableau` diagrams. If you want to use `\aTabSet` to change the sizes of all diagrams and tableaux in your document it may be better to use the `box height` and `box width` options.

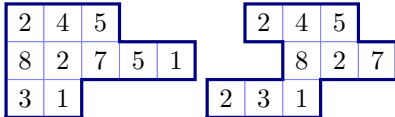


```
\Tableau[starstyle={blue}]{..*, .**, .*}
```



```
\tikzset{R/.style={fill=blue!20,draw=red,thick}}
\Tableau{12[R]3,4[R]5}
\Tableau[starstyle={fill=blue!50,text=white}]{1*23,4*5}
```

Compositions. As we have defined them, tableaux are always of *partition* shape, in that the lengths of the rows form a weakly decreasing sequence. In fact, the `\Tableau` command also accepts tableaux of *composition* shape, where the lengths of the rows are not necessarily in decreasing order



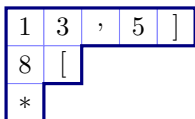
```
\Tableau{ 245,82751,31 }
\SkewTableau{1,2}{ 245,827,231 }
```

Empty rows in tableaux and Young diagrams are not supported!

Drawing order. When `TikZ` constructs a picture, later objects are drawn over the top of earlier ones, which means that earlier features can be obscured or lost. For this reason it is useful to know the drawing order used by :

- First, the tableau entries that use the default styling are placed, in the order that appear in the *tableau specifications*.
- Secondly, the styled tableau entries are placed, in the order that they appear in the *tableau specifications*.
- Next, the nodes in any ribbons are placed, again in the order that the ribbons are listed. For each ribbon, first the border of the ribbon is drawn, together with any styling for the ribbon, and then the boxes in the ribbon, together with any styling, are added.
- The boundary of the tableau is drawn next.
- Finally, the snobs are drawn, in the order that they are specified.

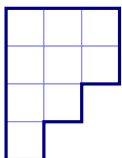
Special characters. The characters `,`, `*` and `[]` each have special meanings in the *tableau specifications*. If you want these characters to be entries in a tableau, then you need to enclose them in braces:



```
\Tableau{ 13{,}5], 8{[}, {*}} }
```

There is no need to enclose `]` in braces because it only becomes special when it is preceded by a `[` character.

1.2. Young diagrams. A *Young diagram*, or simply a *diagram*, is an unlabelled tableau. Diagrams can be drawn using the `\Tableau` command, for example:



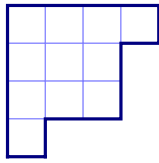
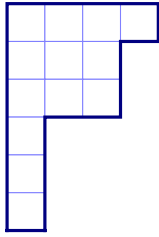
```
\Tableau{~~~,~~~,~~,~}
```

This approach works but is slightly cumbersome, so `aTableau` provides the `\Diagram` command, which is uses almost the same syntax as the `\Tableau` command:

```
\Diagram (x,y) [options] {partition}
```

As with the `\Tableau` command, the (x,y) -coordinates are needed if and only if the diagram is inside a `tikzpicture` environment.

Inside the `\Diagram` command, partitions can be specified as either a comma separated list of weakly decreasing positive integers, or by using exponential notation for partitions:



```
\aTabSet{align=centre}
\Diagram{4,3^2,1^3}
\Diagram{4,3,3,1}
```

Internally, `\Diagram` actually does something like the first example in this section, so all of the options for `\Tableau` can also be used with `\Diagram`, so we do not repeat all of the options in this section. There are also new options that apply specifically to diagrams.

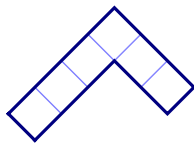
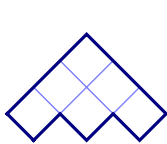
We start by advertising that `\Diagram` supports the four different conventions for diagrams, with `english` being the default:

`english` (default: `english`)

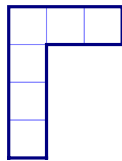
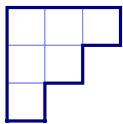
`french`

`ukrainian`

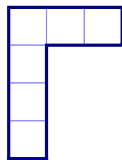
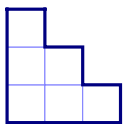
`australian`



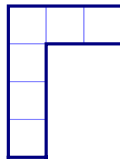
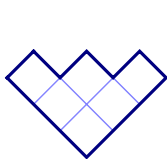
```
\Diagram[australian]{3,2,1}
\Diagram[australian]{3,1^3}
```



```
% English is the default convention
\Diagram[english]{3,2,1}
\Diagram{3,1^3}
```



```
\Diagram[french]{3,2,1}
\Diagram{3,1^3}
```



```
\Diagram[ukrainian]{3,2,1}
\Diagram{3,1^3}
```

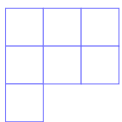
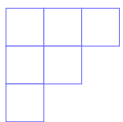
`border` (default: `true`)

[accepts: true/false]

`no border` (default: `false`)

[accepts: true/false]

If `border` is true, then the border of the tableau is drawn. If `border` is false, then the border of the tableau is not drawn. By default, the border is drawn. The `no border` option, which is provided only for completeness, is the inverse of `border`. That is, `no border` is equivalent to `border=false`, and `no border=false` is equivalent to `border=true`.

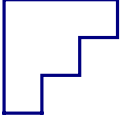


```
\Diagram[border=false]{3,2,1}
\Diagram[no border]{3^2,1}
```

`border only` (default: `false`)

[accepts: false/true]

If `border only` is true, then only the border of the tableau is drawn, and not the internal walls around the boxes.

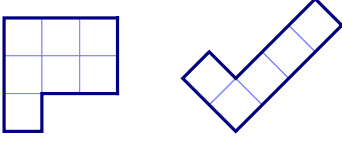


ribbons

[accepts: ribbon specification]

snobs

[accepts: ribbon specification]



```
\Diagram[conjugate]{3,2,2}
\Diagram[ukrainian, conjugate]{2,1^3}
```

conjugate (default: **false**)

[accepts: false/true]

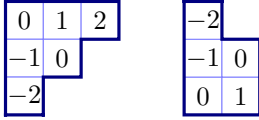
Draws the conjugate diagram, which has the rows and columns interchanged.

show

[accepts: contents,final,hooks,initial,residue]

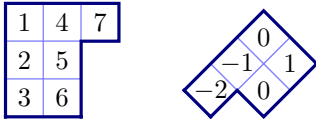
Use **show** option to draw particular types of tableau of the specified shape. The possible choices are:

- **show=contents** print the diagram where each box is labelled by its *content*, which is the row index minus the column index.



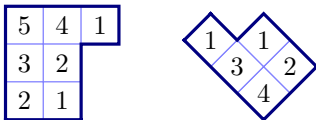
```
\Diagram[show=contents]{3,2,1}
\Diagram[french, show=contents]{2^2,1}
```

- **show=final** print the *final tableau* of this shape, which has the numbers $1, 2, \dots, n$ entered in order from top to bottom down successive columns (with appropriate modifications for the non-English conventions).



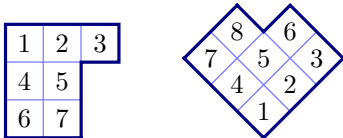
```
\Diagram[show=final]{3,2,2}
\Diagram[australian, show=contents]{2^2,1}
```

- **show=hooks** print the diagram where each box is labelled by the corresponding *hook length*. If λ is a partition and λ' is its conjugate partition then the hook length of the box in row i and column j is $\lambda_i - i + \lambda'_j - j + 1$.



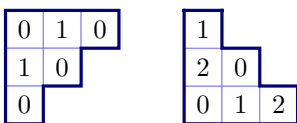
```
\Diagram[show=hooks]{3,2,2}
\Diagram[ukrainian, show=hooks]{2^2,1}
```

- **show=initial** print the *initial tableau* of this shape, which has the numbers $1, 2, \dots, n$ entered in order from top left to right along down successive rows (with appropriate modifications for the non-English conventions).



```
\Diagram[show=initial]{3,2,2}
\Diagram[ukrainian, show=initial]{3^2,2}
```

- **show=residues** print the diagram where each box is labelled by its *residue*, which is the row index minus the column index modulo some integer e , which must also be supplied.



```
\Diagram[show=residues, e=2]{3,2,1}
\Diagram[french, show=residues, e=3]{3,2,1}
```

1.3. Skew tableau and skew diagrams. A partition μ is *contained* in a partition λ , which is written as $\mu \subseteq \lambda$, if $\mu_k \leq \lambda_k$, for $k \geq 0$. If $\mu \subseteq \lambda$ then the *skew partition* $\lambda/\mu = \{(r, c) \in \lambda \mid (r, c) \notin \mu\}$ is the set of nodes that are in λ and not in μ . In this manual, we call μ the *inner shape* and λ/μ the *outer shape*.

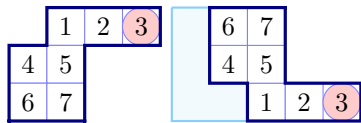
A *skew tableau* is a labelling of the nodes in the diagram of a skew partition. Skew partitions and skew tableau can be drawn using the commands:

```
\SkewDiagram (x,y) [options] {inner shape} {outer shape}
```

```
\SkewTableau (x,y) [options] {inner shape} {skew tableau}
```

As with the previous commands, the (x,y) -coordinates should be given if and only if the picture is inside a `tikzpicture` environment. The inner and outer shapes can either be given as a comma separated list of integers or in exponential notation. In a `\SkewTableau`, the tableau entries are specified exactly as in the `\Tableau` command. In particular, they can be given optional style commands.

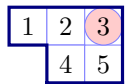
All of the options for the `\Diagram` command can be used with the `\SkewDiagram` command and, similarly, all of the options for the `\Tableau` command can be used with the `\SkewTableau` command.



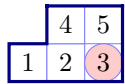
```
\tikzset{R/.style={fill=red!20,circle}}
\SkewTableau{2,1^2}{12[R]3,45,67}
\SkewTableau[french,skewwall]{2,1^2}{12[R]3,45,67}
```

1.4. Shifted tableau and shifted diagrams. `\ShiftedDiagram(x,y) [options] {partition }`

```
\ShiftedTableau (x,y) [options] {partition }
```



```
\tikzset{R/.style={fill=red!20,circle}}
\ShiftedTableau{12[R]3,45}
```

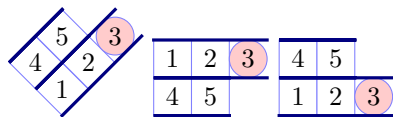


```
\tikzset{R/.style={fill=red!20,circle}}
\ShiftedTableau[french]{12[R]3,45}
```

1.5. Tabloids. A *tabloid* is an equivalence class of tableau where two tableaux are equivalent if they have the same entries in each row, up to a permutation. In the literature, tabloids are usually drawn with lines above and below each row, and no side boundary walls

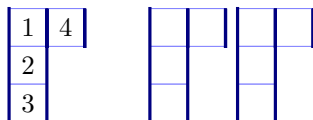
```
\Tabloid (x,y) [options] {partition }
```

The `\Tabloid` command functions in the same way as the `\Tableau` command, except that it draws tabloids.



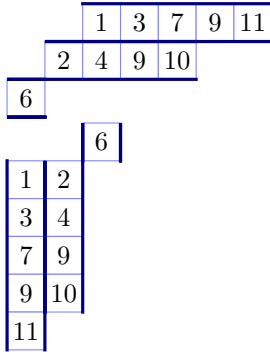
```
\tikzset{R/.style={fill=red!20,circle,minimum size=5mm}}
\Tabloid[ukrainian]{12[R]3,45}
\Tabloid[english]{12[R]3,45}
\Tabloid[french]{12[R]3,45}
```

Some authors work with *column tabloids*. Such tableaux do not have a dedicated `aTableau` command, however, they can be drawn using the `conjugate` option.



```
\Tabloid[conjugate]{123,4}
\Tabloid[conjugate]{~~~,~}
\Diagram[conjugate,tabloid]{3,1}
```

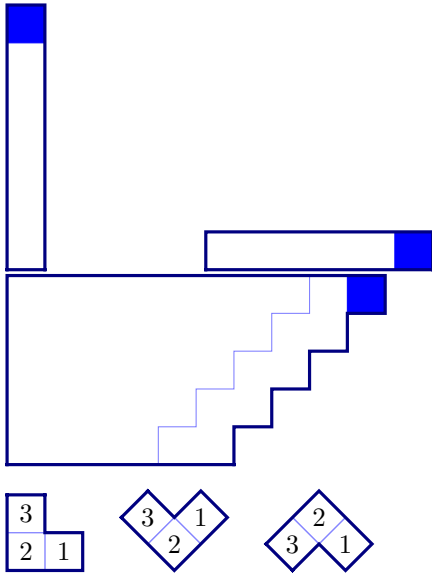
Similarly, skew tabloids can be drawn using the `skew` option.



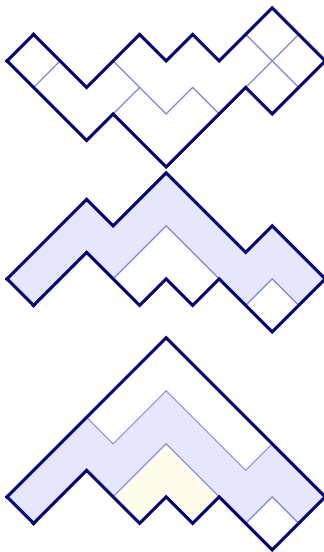
```
\aTabSet{align=center}
\Tabloid[skew={2,1}]{1379{11},249{10},6,}
\Tabloid[conjugate,skew={1^2}]{1379{11},249{10},6,}
```

1.6. **Ribbon tableaux.** A *ribbon* in a tableau, or in a diagram, is a connected strip of nodes that does not contain a (2×2) -square . The *head* of a ribbon R is the unique node $(r, c) \in R$ such that $r' > r$ and $c' < c$ whenever $(r', c') \in R$ and $(r', c') \neq (r, c)$.

```
\RibbonTableau (x,y) [options] {ribbons}
```



```
\RibbonTableau{[blue]11rrrrrr}
\RibbonTableau{[blue]16ccccc}
\RibbonTableau{[blue]1{10}crrcrrcrrc}
```



```
\RibbonTableau[french] { 12_1 c_2 r_3 }
\RibbonTableau[ukrainian] { 12_1 c_2 r_3 }
\RibbonTableau[australian]{ 12_1 c_2 r_3 }
```

```
\RibbonTableau[skew={4,1^2},ukrainian]
{15,16,26,25crrcrr,23cr,42cr,61}
\RibbonTableau[skew={4,1^2},australian]
{([fill=blue!10]16crrcrrcrr,26,34cr}
\RibbonTableau[australian]
{([fill=blue!10]16crrcrrcrr,26,([fill=yellow!10]34cr}
```

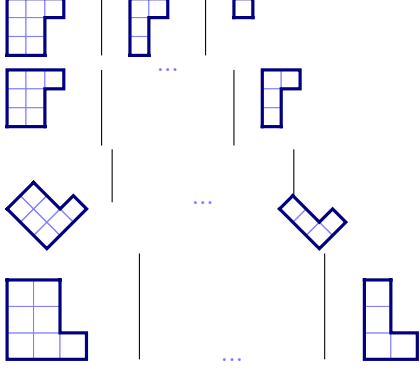


```
\RibbonTableau{
[fill=blue!20]12_1 [fill=red!20]c_2
[fill=yellow!20]r_3 }
```

1.7. Multidiagrams and multitableaux. Multitableaux and multidiagrams, with are ℓ -tuples of tableaux diagrams respectively, appear in the representation theory of wreath products and cyclotomic Hecke algebras. It is not difficult to construct such tableaux and diagrams using the `\Tableau` and `\Diagram` commands. For convenience, `aTableau` provides the `\Multidiagram` and `\Multitableau` commands for drawing these diagrams.

`\Multidiagram (x,y) [options] {multipartition}`

`\Multitableau (x,y) [options] {multitableau}`

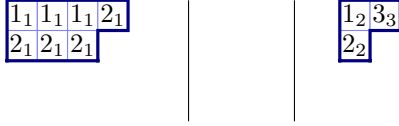


`\Multidiagram[scale=0.5]{3,2^2||2,1^2||1}`

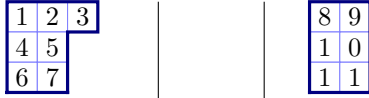
`\Multidiagram[scale=0.5,rows=3,rows=3]{3,2^2||...||2,1^2}`

`\Multidiagram[ukrainian,scale=0.5,rows=3,cols=3]{3,2^2||...||2,1^2}`

`\Multidiagram[french,scale=0.7,rows=3]{3,2^2||...||2,1^2}`



`\Multitableau[rows=3,scale=0.8,font=\tiny]`
`{{1_1}{1_1}{1_1}{2_1},{2_1}{2_1}{2_1} ||`
`{1_2}{3_3},{2_2} ||`
`{1_3}{1_3},{2_3}}`



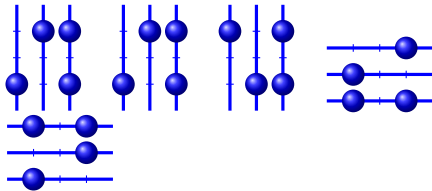
`\Multitableau[rows=3,scale=0.8,font=\scriptsize]`
`{123,45,67||89,{10},{11}||...||{12}}`



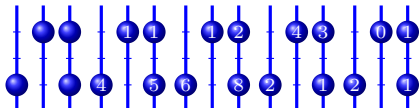
2. ABACUSES

An abacus

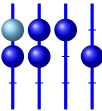
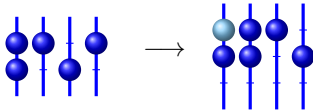
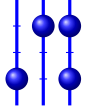
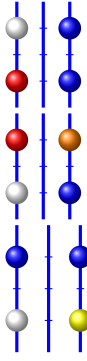
`\Abacus (x,y) [options] {number of runners} {bead specifications}`



`\Abacus{3}{5,4,1,1}`
`\Abacus[south]{3}{5,4,1,1}`
`\Abacus[north]{3}{5,4,1,1}`
`\Abacus[east]{3}{5,4,1,1}`
`\Abacus[west]{3}{5,4,1,1}`



`\Abacus{3}{5,4,1,1}`
`\Abacus[show=shape,text=yellow]{3}{5,4,1,1}`
`\Abacus[show=beta,text=red]{3}{5,4,1,1}`
`\Abacus[show=rows]{3}{5,4,1,1}`
`\Abacus[show=residue]{3}{5,4,1,1}`



```
\tikzset{R/.style={ball color=red},
          W/.style={ball color=white},
          O/.style={ball color=orange}
}

\Abacus{3}{5,[R]4,1^2,[W]0^4}

\Abacus[north]{3}{[O]5^2_{ooo},[R]4,1^2,[W]0^4}

\Abacus[north,scale=1.2,starstyle={ball
color=yellow}]{3}{5,4,*1^2,[W]0^4}
```

```
\Abacus[text=white,font=\tiny]{3}{5_a,4_b,1_c,1_d}
```

```
\aTabSet{align=centre}
\tikzset{B/.style={ball color=LightSkyBlue}}
\Abacus[beta numbers]{4}{6,4,3,1,0}
$\longrightarrow$
\Abacus[beta numbers]{4}{7,5,4,2,1,[B]0}
```

3. DYNKIN QUIVERS?

4. BRAID DIAGRAMS AND KLRW DIAGRAMS?

5. `\Tableau` SETTINGS

6. THE CODE

Every command provided by `aTableau` uses `TikZ`, with the bulk of the code in the package being written in $\text{\LaTeX}$ 3. Most of the tableau commands are special cases of the `\Tableau` command. Here is a dictionary:

Command	Equivalent <code>\Tableau</code> command
<code>\ShiftedTableau{...}</code>	<code>\Tableau[shifted]{...}</code>
<code>\SkewTableau{μ}{...}</code>	<code>\Tableau[skew=μ]{...}</code>
<code>\Tabloid{...}</code>	<code>\Tableau[tabloid]{...}</code>

The `\Diagram` command also invokes the `\Tableau` command, however, this requires some involved internal trickery. For example, `\Diagram{3,2^2}` is the same as `\Tableau{...,...,...}`.

Use the `\EAbacus` command to typeset an abacus

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